

Intelligent Recruitment and Talent Analytics Platform

*A Project Based Learning Report Submitted in partial fulfilment of the requirements for the
award of the degree*

of

Bachelor of Technology

in The Department of Artificial Intelligence and Data Science

**Database Systems Engineering and Distributed Backend Development
25CS1302E**

Submitted by

2510080029 - Oddesar Sudeeptha

2510080037 - Lokrey Latika

2510080043 - Samineni Sai Siri

2510080061 - Muvva Kovidha

Under the guidance of

Dr.Laiphangbam Melinda



Department of Electronics and Communication Engineering

Koneru Lakshmaiah Education Foundation, Aziz Nagar

Aziz Nagar – 500075

July - 2026

ABSTRACT

The Intelligent Recruitment and Talent Analytics Platform is a web-based application developed to automate and streamline the recruitment process. Traditional recruitment systems often involve manual candidate screening, job posting management, and interview scheduling, which can be time-consuming and prone to errors. Modern recruitment platforms leverage web technologies and database systems to improve efficiency and accuracy in hiring processes [1].

The proposed system provides a centralized platform where candidates can register, apply for jobs, and track recruitment activities, while recruiters can manage job postings, review candidate applications, schedule interviews, and analyze recruitment performance. The platform incorporates a Talent Matching module that compares candidate skills with job requirements to support effective shortlisting and hiring decisions [2].

The application is implemented using HTML, CSS, Java Servlets, Apache Tomcat, JDBC, and PostgreSQL [3][4][5]. Recruitment data is stored and managed efficiently through a relational database system, while the analytics dashboard provides insights into recruitment performance and interview outcomes. The system improves recruitment efficiency, reduces manual effort, and supports data-driven hiring decisions.

LIST OF FIGURES

Figure No.	Figure Title
Figure 2.1	Workflow of Intelligent Recruitment and Talent Analytics Platform
Figure 2.2	System Architecture of Intelligent Recruitment and Talent Analytics Platform
Figure 2.3	Use Case Diagram of the Intelligent Recruitment and Talent Analytics Platform
Figure 2.4	Level-0 Data Flow Diagram
Figure 3.1	Home Page of Intelligent Recruitment Platform
Figure 3.2	Candidate Registration Module
Figure 3.3	Recruiter Login Module
Figure 3.4	Job Management Module
Figure 3.5	View Jobs Module
Figure 3.6	Job Application Module
Figure 3.7	Interview Scheduling Module
Figure 3.8	Talent Matching Module
Figure 3.9	Recruitment Analytics Dashboard
Figure 3.10	Java Servlet to PostgreSQL Database Connectivity Flow Using JDBC

Figure 3.11	Candidate–Job Skill Matching and Matching Information Calculation Process
Figure 3.12	Recruitment Data Processing and Analytics Dashboard Generation Flow
Figure 4.1	Candidate Table in PostgreSQL Database
Figure 4.2	Job Table in PostgreSQL Database
Figure 4.3	Application Table in PostgreSQL Database
Figure 4.4	Interview Table in PostgreSQL Database
Figure 4.5	Recruiter Table in PostgreSQL Database
Figure 4.6(a)	Candidate–Application–Job Relationship
Figure 4.6(b)	Candidate–Interview–Job Relationship
Figure 4.7	Entity Relationship Diagram of Candidate, Application, Interview, Job, and Recruiter
Figure 5.1	Recruitment Analytics Output

LIST OF TABLES

Table No.	Table Title
Table 1.1	Project Modules
Table 3.1	Software Requirements
Table 3.2	Hardware Requirements
Table 4.1	Candidate Table Structure
Table 4.2	Job Table Structure
Table 4.3	Application Table Structure
Table 4.4	Interview Table Structure
Table 4.5	Recruiter Table Structure
Table 5.1	Testing Results
Table 5.2	Functional Testing Results of the Intelligent Recruitment and Talent Analytics Platform

TABLE OF CONTENTS

Section	Page
ABSTRACT	2
LIST OF FIGURES	3
LIST OF TABLES	4
TABLE OF CONTENTS	5–7
CHAPTER 1: INTRODUCTION	8
1.1 Objectives	9
1.2 Problem Statement	9
1.3 Existing System	10
1.4 Proposed System	10
1.5 Scope of the Project	11
1.6 Project Modules	12
CHAPTER 2: METHODOLOGY	13
2.1 System Workflow	13
2.2 System Architecture	14
2.3 Functional Requirements	15
2.4 Non-Functional Requirements	16
2.5 Candidate Management Module	17
2.6 Job Management Module	17
2.7 Job Application Module	17
2.8 Talent Matching Module	17
2.9 Interview Scheduling Module	18

2.10 Analytics Dashboard Module	18
2.11 Use Case Diagram	18
2.12 Data Flow Diagram	19
2.13 Module Interaction	19
CHAPTER 3: IMPLEMENTATION	21
3.1 Software Requirements	21
3.2 Hardware Requirements	21
3.3 Home Page	22
3.4 Candidate Registration Module	22
3.5 Recruiter Login Module	23
3.6 Job Management Module	23
3.7 View Jobs Module	24
3.8 Job Application Module	24–25
3.9 Interview Scheduling Module	25–26
3.10 Talent Matching Module	26
3.11 Analytics Dashboard	27
3.12 Database Connectivity	27–28
3.13 Backend Implementation	29
3.14 Frontend Implementation	30–31
3.15 Talent Matching Algorithm	32
3.16 Analytics Implementation	33–34
CHAPTER 4: DATABASE DESIGN	35
4.1 Candidate Table	35
4.2 Job Table	36
4.3 Application Table	37

4.4 Interview Table	38
4.5 Recruiter Table	39
4.6 Database Relationships	40
4.7 Entity Relationship Diagram	41–42
4.8 Database Normalization	43
4.9 Database Connectivity	44
CHAPTER 5: RESULTS AND DISCUSSION	45
5.1 Results Obtained	45
5.2 Discussion	46
5.3 Advantages	47
5.4 Testing Results	47
5.5 Functional Testing	48
5.6 Database Testing	49
5.7 User Interface Testing	50
5.8 Performance Discussion	50
5.9 Limitations	51
CHAPTER 6: CONCLUSION AND FUTURE SCOPE	52
6.1 Conclusion	52
6.2 Future Scope	52
6.3 Final Project Outcome	53
CHAPTER 7: REFERENCES	54

INTELLIGENT RECRUITMENT AND TALENT ANALYTICS PLATFORM

CHAPTER 1: INTRODUCTION

Recruitment is a critical organizational activity that directly impacts workforce quality and organizational performance. As the number of job applicants increases, organizations require efficient systems to manage candidate information, job postings, interview schedules, and hiring decisions. Traditional recruitment approaches often rely on spreadsheets, email communication, and manual record keeping, resulting in inefficiencies and increased administrative effort [8].

The Intelligent Recruitment and Talent Analytics Platform addresses these challenges by providing an integrated recruitment management system. The platform allows candidates to create profiles, recruiters to manage vacancies, and organizations to monitor recruitment activities through analytical reports. The integration of web technologies and database management systems provides a reliable and scalable recruitment solution [7].

The project also incorporates talent analytics concepts, which enable organizations to evaluate candidate-job compatibility through skill-based matching. This improves candidate selection and helps recruiters identify qualified applicants more efficiently [2].

1.1 OBJECTIVES

The major objectives of the project are:

1. To automate candidate registration and profile management.
2. To provide job posting and recruitment management facilities.
3. To enable candidates to apply for available jobs online.
4. To automate interview scheduling and tracking.
5. To implement skill-based talent matching.
6. To generate recruitment analytics and performance reports.
7. To maintain recruitment information securely using PostgreSQL.
8. To provide a user-friendly web interface for candidates and recruiters.

1.2 PROBLEM STATEMENT

Traditional recruitment processes involve several manual activities such as collecting candidate information, maintaining job postings, reviewing applications, scheduling interviews, and tracking recruitment progress. When these activities are handled through spreadsheets, emails, and separate records, recruiters may face difficulties in maintaining accurate and up-to-date information.

Another challenge is identifying suitable candidates from a large number of applications. Manual comparison of candidate skills with job requirements can be time-consuming and may result in inconsistent shortlisting decisions. In addition, organizations may not have a centralized mechanism to monitor recruitment statistics and interview outcomes.

Therefore, there is a need for an integrated recruitment platform that can centralize candidate information, job postings, applications, interviews, talent matching, and

recruitment analytics. The proposed Intelligent Recruitment and Talent Analytics Platform addresses these requirements through a web-based application integrated with a relational database.

1.3 EXISTING SYSTEM

In a conventional recruitment system, candidate information is frequently collected through application forms, emails, spreadsheets, or separate recruitment portals. Recruiters manually review candidate profiles and compare their skills with job requirements.

The major limitations of the existing approach are:

- Manual candidate information management.
- Difficulty in maintaining large numbers of applications.
- Time-consuming candidate screening.
- Manual interview scheduling.
- Limited recruitment performance monitoring.
- Lack of centralized recruitment information.
- Difficulty in comparing candidate skills with job requirements.
- Increased possibility of human errors.

These limitations demonstrate the need for an integrated recruitment management solution.

1.4 PROPOSED SYSTEM

The proposed Intelligent Recruitment and Talent Analytics Platform provides a centralized web-based environment for managing recruitment activities.

The system allows candidates to register their profiles, browse available jobs, and submit applications. Recruiters can create job postings, review candidate applications, schedule interviews, and monitor recruitment statistics.

The Talent Matching module compares candidate skills with the skills required for a particular job and generates a matching result. The Analytics Dashboard provides recruitment statistics that help recruiters understand the current status of recruitment activities.

The proposed system therefore reduces manual effort and provides an organized workflow from candidate registration to recruitment analytics.

1.5 SCOPE OF THE PROJECT

The scope of the project includes the development of a web-based recruitment management platform that supports the major stages of the recruitment process.

The system provides candidate registration, recruiter authentication, job management, job applications, talent matching, interview scheduling, and recruitment analytics.

The platform maintains recruitment data using PostgreSQL and uses Java Servlets for backend processing. JDBC provides communication between the Java application and the database.

The system can be used as a foundation for future recruitment solutions that incorporate artificial intelligence, machine learning, resume parsing, automated notifications, and cloud deployment.

1.6 PROJECT MODULES

Module	Description
Candidate Management	Candidate Registration and Profile Management
Job Management	Create and Manage Job Openings
Job Application	Apply for Available Jobs
Interview Scheduling	Schedule Candidate Interviews
Talent Matching	Match Candidate Skills with Job Requirements
Analytics Dashboard	View Recruitment Statistics
Authentication	Recruiter Login System

Table 1.1: Project Modules

CHAPTER 2: METHODOLOGY

The Intelligent Recruitment and Talent Analytics Platform follows a modular architecture that separates different recruitment functions into independent modules. The system uses Java Servlets for backend processing, PostgreSQL for database management, and Apache Tomcat as the deployment server [3][4][5].

2.1 SYSTEM WORKFLOW



Figure 2.1: Workflow of Intelligent Recruitment and Talent Analytics Platform

2.2 SYSTEM ARCHITECTURE

The system follows a three-tier architecture consisting of presentation, application, and database layers. The frontend is developed using HTML and CSS. Java Servlets process user requests and implement business logic. PostgreSQL stores candidate, job, application, and interview data. The Analytics Dashboard retrieves information from the database and displays recruitment statistics.

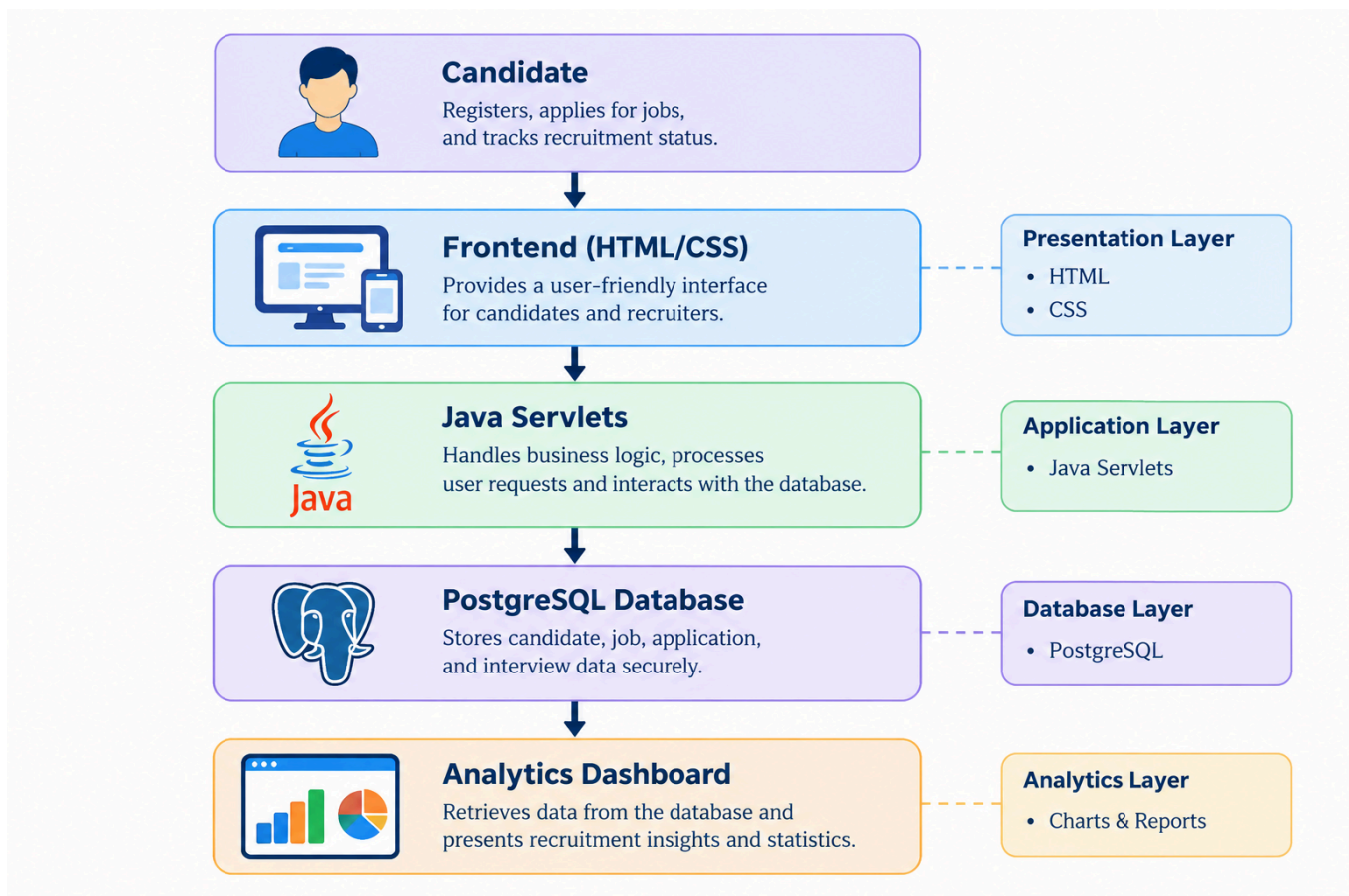


Figure 2.2: System Architecture of Intelligent Recruitment and Talent Analytics Platform

2.3 FUNCTIONAL REQUIREMENTS

The functional requirements describe the operations that the system must perform.

Candidate Management

The system shall allow candidates to register by entering their personal information, contact details, qualifications, and skills.

Recruiter Authentication

The system shall allow authorized recruiters to log into the platform and access recruitment management functions.

Job Management

Recruiters shall be able to create and manage job postings containing job title, description, skills, experience, and other relevant information.

Job Application

Candidates shall be able to browse available jobs and submit applications.

Talent Matching

The system shall compare candidate skills with required job skills and generate a matching result.

Interview Scheduling

Recruiters shall be able to schedule interviews and maintain interview status.

Analytics

The system shall display recruitment statistics through the Analytics Dashboard.

2.4 NON-FUNCTIONAL REQUIREMENTS

Performance

The system should respond to user requests within a reasonable amount of time.

Usability

The interface should be simple and understandable for both candidates and recruiters.

Reliability

The system should maintain recruitment records accurately and consistently.

Security

Authentication should prevent unauthorized users from accessing recruiter functionality.

Maintainability

The modular architecture should make it easier to modify or extend individual components.

Scalability

The system architecture should allow additional recruitment features to be incorporated in future versions.

2.5 CANDIDATE MANAGEMENT MODULE

The Candidate Management module allows job seekers to register their profiles by entering personal information, contact details, educational qualifications, and technical skills. The information is validated and stored in the PostgreSQL database using JDBC connectivity [4]. This module serves as the foundation for subsequent recruitment activities.

2.6 JOB MANAGEMENT MODULE

Recruiters can create and manage job openings by specifying job titles, descriptions, required qualifications, experience levels, and skill requirements. This module helps organizations maintain an organized repository of available opportunities and simplifies vacancy management [1].

2.7 JOB APPLICATION MODULE

Candidates can browse available job opportunities and submit applications directly through the platform. Application details are stored in the database and linked to corresponding job postings, enabling recruiters to review candidate submissions efficiently.

2.8 TALENT MATCHING MODULE

Talent matching is a key feature of the platform. The system compares candidate skills against job requirements and generates a matching score. This assists recruiters in identifying candidates who best fit a particular position and supports data-driven recruitment decisions [2].

2.9 INTERVIEW SCHEDULING MODULE

The Interview Scheduling module allows recruiters to schedule candidate interviews, assign interview dates and times, and update interview status. Interview records are stored in the database, enabling recruiters to track recruitment progress effectively [10].

2.10 ANALYTICS DASHBOARD MODULE

The Analytics Dashboard provides recruitment-related statistics such as total candidates, total jobs, total interviews, completed interviews, and cancelled interviews. These analytical insights support performance monitoring and strategic recruitment planning [2].

2.11 USE CASE DIAGRAM

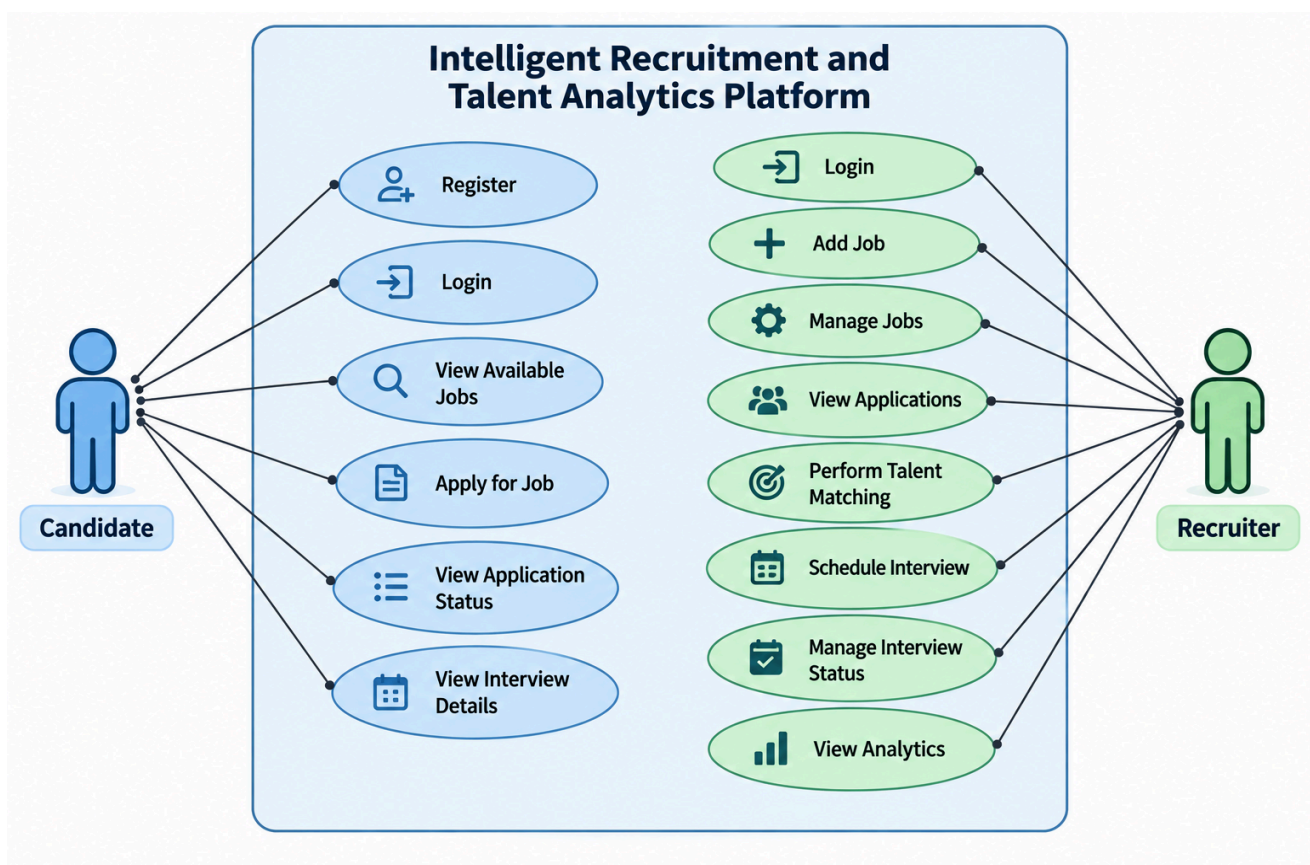


Figure 2.3: Use Case Diagram of the Intelligent Recruitment and Talent Analytics Platform

2.12 DATA FLOW DIAGRAM

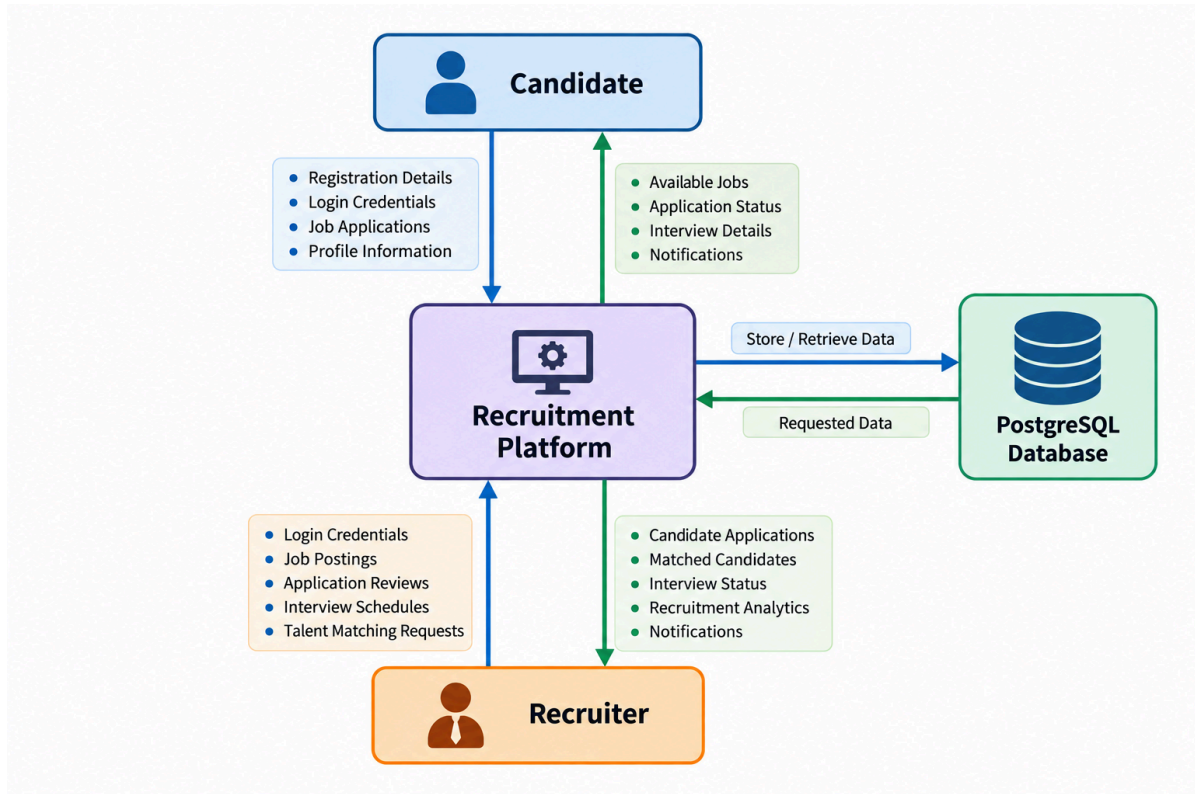


Figure 2.4: Level-0 Data Flow Diagram

2.13 MODULE INTERACTION

The modules of the proposed platform are interconnected to provide a complete recruitment workflow. Candidate Registration creates candidate profiles that are stored in the database. Job Management allows recruiters to create available vacancies. Candidates can then apply for suitable jobs through the Job Application module.

The Talent Matching module uses candidate skills and job requirements to identify suitable candidates. Shortlisted candidates can be scheduled for interviews through the Interview Scheduling module. Finally, recruitment information stored in the database is used by the Analytics Dashboard to generate recruitment statistics.

CHAPTER 3: IMPLEMENTATION

3.1 SOFTWARE REQUIREMENTS

S.No	Software	Purpose
1	Java Development Kit (JDK)	Java Application Development
2	Eclipse IDE	Coding and Project Development
3	Apache Tomcat Server	Deployment and Execution of Web Application
4	PostgreSQL	Database Management System
5	pgAdmin	Database Administration Tool
6	HTML	Frontend Structure Design
7	CSS	User Interface Styling
8	JDBC Driver	Database Connectivity

Table 3.1: Software Requirements

3.2 HARDWARE REQUIREMENTS

S.No	Component	Specification
1	Processor	Intel Core i5 or Higher
2	RAM	Minimum 8 GB
3	Storage	256 GB or Higher
4	Operating System	Windows 11

Table 3.2: Hardware Requirements

3.3 HOME PAGE

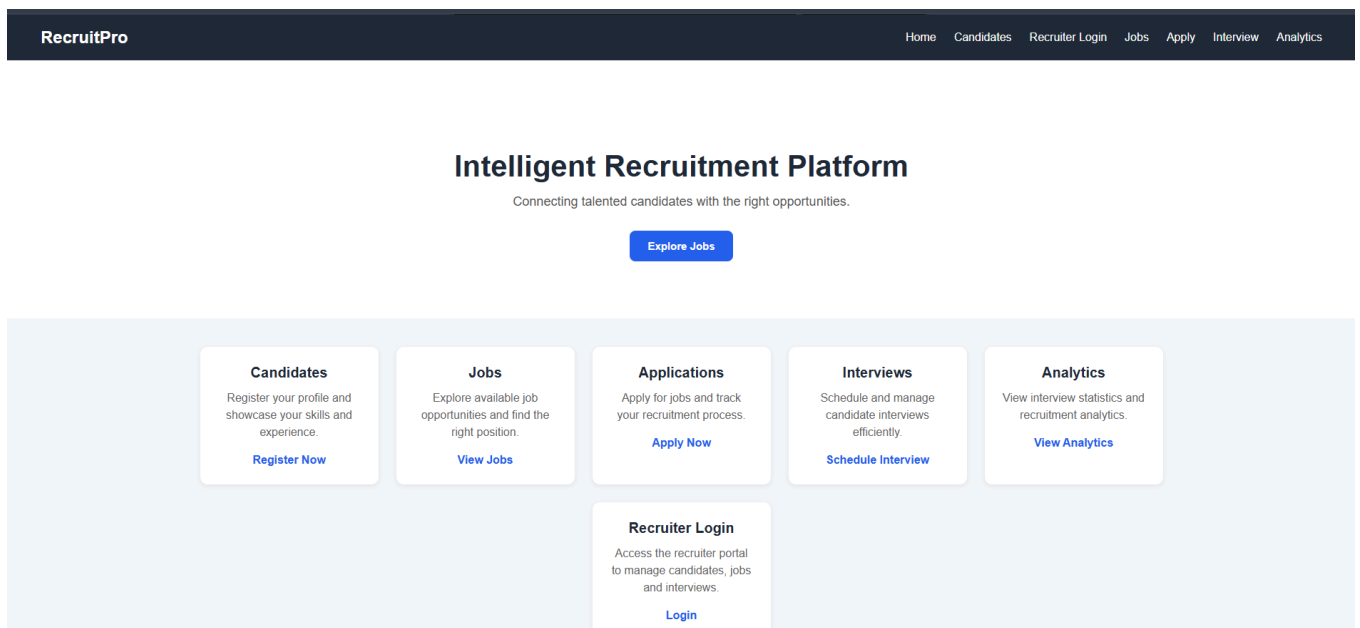


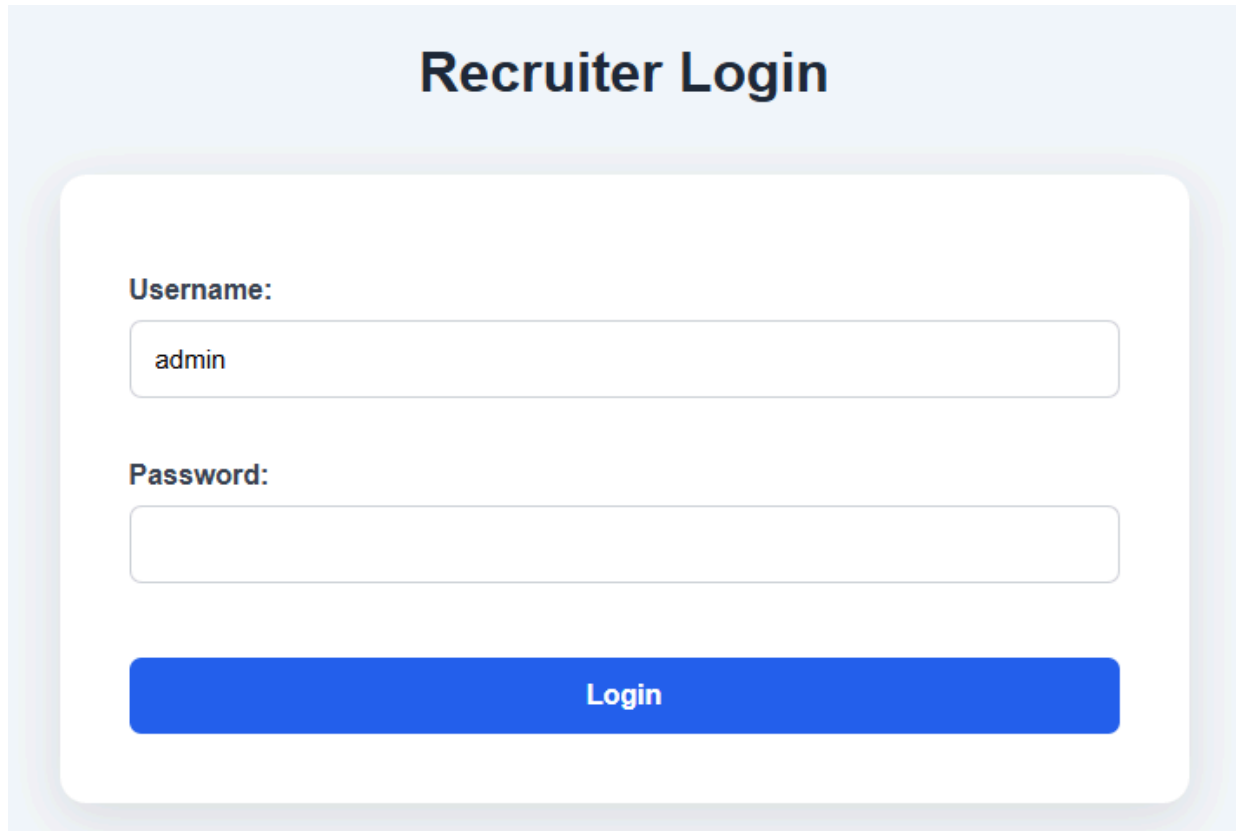
Figure 3.1: Home Page of Intelligent Recruitment Platform

3.4 CANDIDATE REGISTRATION MODULE

The image displays the 'Candidate Registration' form. It has a title 'Candidate Registration' at the top. Below the title are four labeled input fields: 'Name:', 'Email:', 'Phone:', and 'Skills:'. Each label is followed by a white rectangular input box. At the bottom of the form is a large blue button with the text 'Register Candidate' in white.

Figure 3.2: Candidate Registration Module

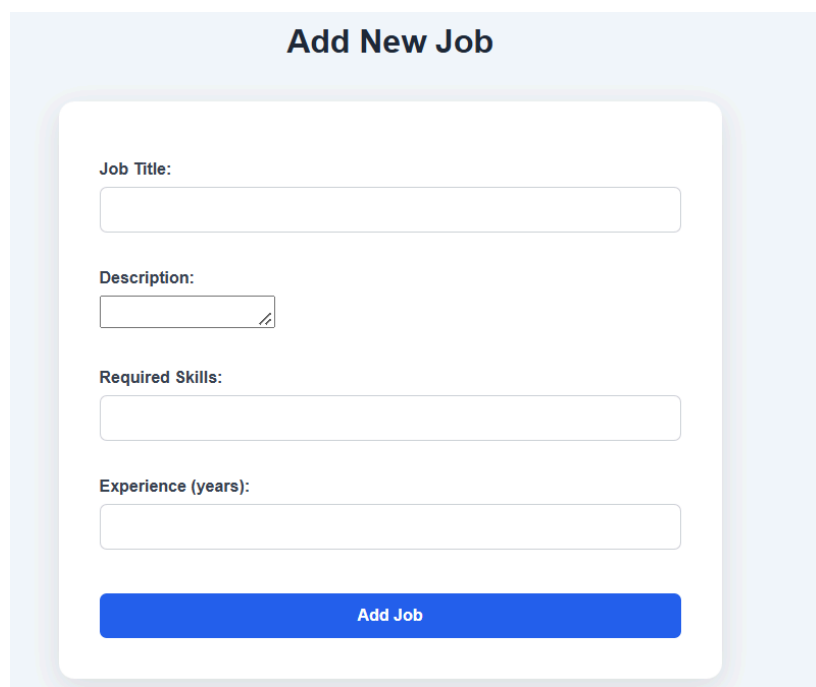
3.5 RECRUITER LOGIN MODULE



The Recruiter Login Module is a web form titled "Recruiter Login". It features a light blue background with a white rounded rectangle in the center. Inside the rectangle, there are two input fields: "Username:" with the text "admin" and "Password:". Below these fields is a blue button labeled "Login".

Figure 3.3: Recruiter Login Module

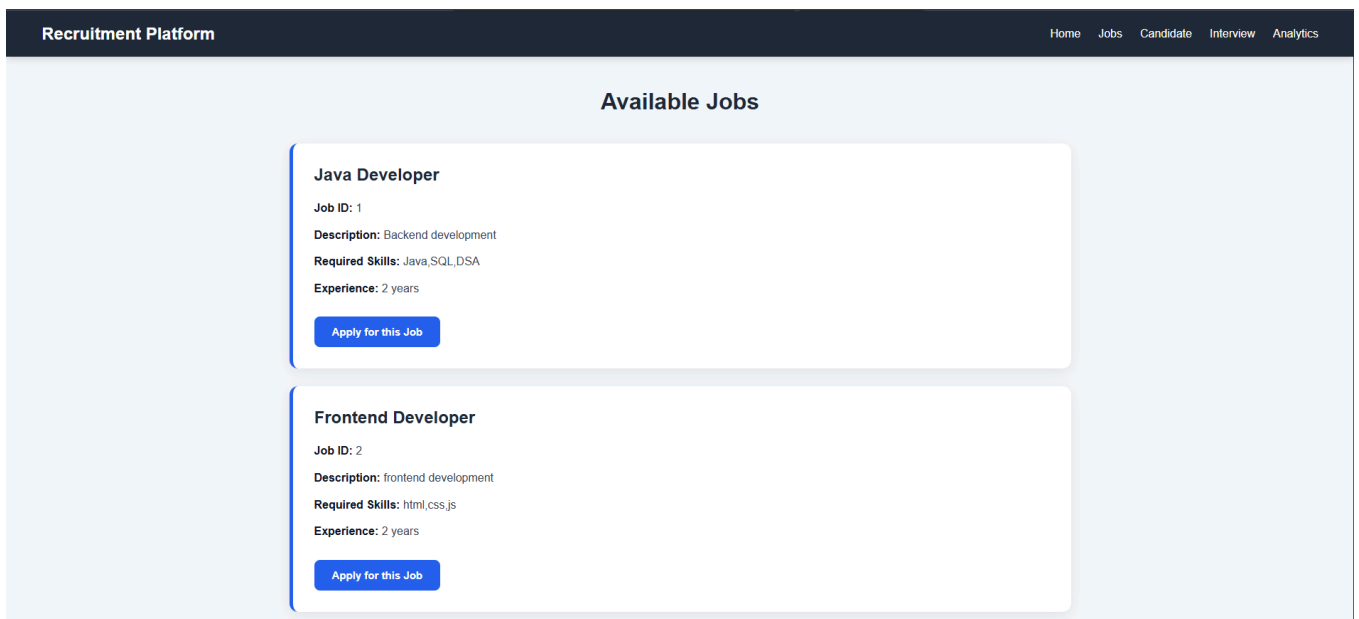
3.6 JOB MANAGEMENT MODULE



The Job Management Module is a web form titled "Add New Job". It features a light blue background with a white rounded rectangle in the center. Inside the rectangle, there are four input fields: "Job Title:", "Description:" (with a text area icon), "Required Skills:", and "Experience (years):". Below these fields is a blue button labeled "Add Job".

Figure 3.4: Job Management Module

3.7 VIEW JOBS MODULE



The screenshot shows a web application titled "Recruitment Platform" with a dark blue header. The header contains navigation links: Home, Jobs, Candidate, Interview, and Analytics. The main content area is titled "Available Jobs" and displays two job listings. Each listing is a white card with a blue border. The first card is for a "Java Developer" position (Job ID: 1) with a description of "Backend development", required skills of "Java,SQL,DSA", and 2 years of experience. The second card is for a "Frontend Developer" position (Job ID: 2) with a description of "frontend development", required skills of "html,css,js", and 2 years of experience. Both cards have a blue "Apply for this Job" button at the bottom.

Recruitment Platform

Home Jobs Candidate Interview Analytics

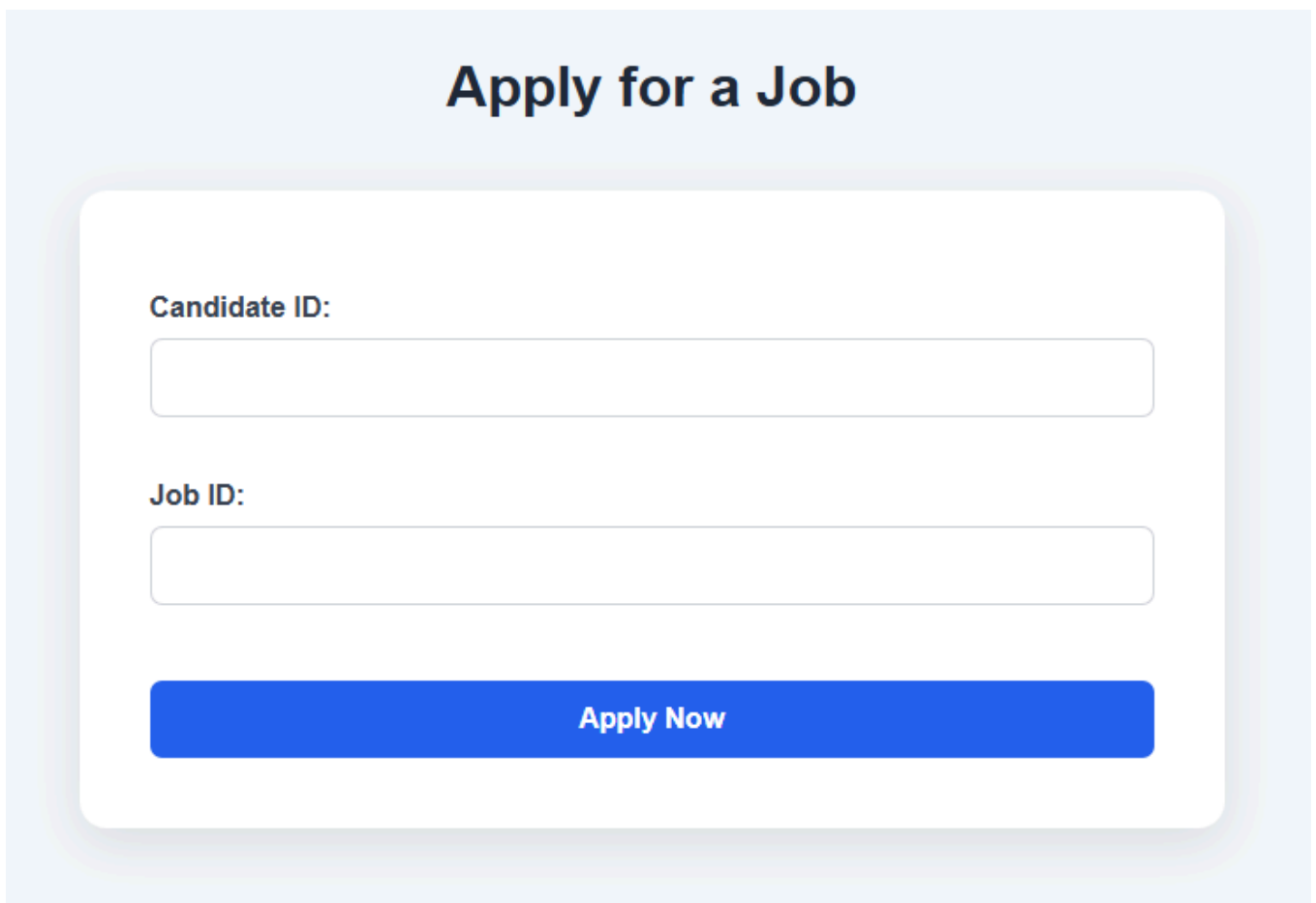
Available Jobs

Java Developer
Job ID: 1
Description: Backend development
Required Skills: Java,SQL,DSA
Experience: 2 years
[Apply for this Job](#)

Frontend Developer
Job ID: 2
Description: frontend development
Required Skills: html,css,js
Experience: 2 years
[Apply for this Job](#)

Figure 3.5: View Jobs Module

3.8 JOB APPLICATION MODULE



The screenshot shows a web application titled "Apply for a Job". The main content area is a white card with a blue border. It contains two input fields: "Candidate ID:" and "Job ID:". Below the input fields is a large blue button labeled "Apply Now".

Apply for a Job

Candidate ID:

Job ID:

[Apply Now](#)

Figure 3.6: Job Application Module

3.9 INTERVIEW SCHEDULING MODULE

Schedule Interview

Application ID:

Interview Date:

dd-mm-yyyy

Interview Time:

--:--

Interview Mode:

Online

Interviewer:

Schedule Interview

Figure 3.7: Interview Scheduling Module

3.10 TALENT MATCHING MODULE

Talent Matching

Select a candidate and job to automatically calculate the skill compatibility score.

Candidate ID:

Job ID:

Find Match

Figure 3.8: Talent Matching Module

3.11 ANALYTICS DASHBOARD

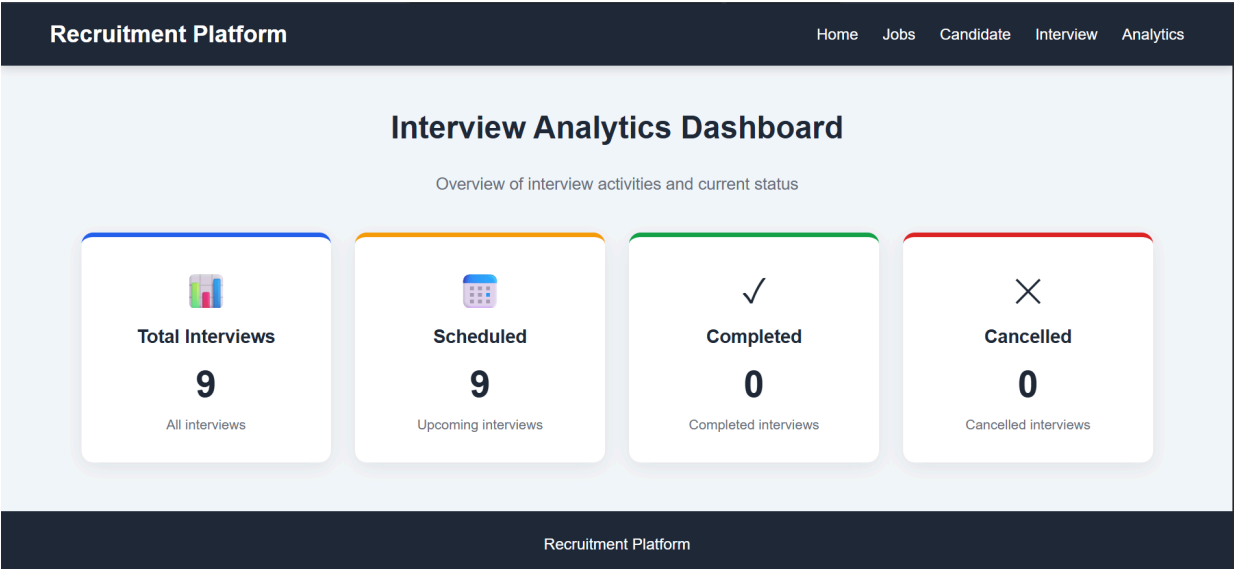


Figure 3.9: Recruitment Analytics Dashboard

3.12 DATABASE CONNECTIVITY

The system uses JDBC to establish communication between Java Servlets and the PostgreSQL database. JDBC allows the application to connect to the database, execute SQL queries, insert records, retrieve information, and update existing records.

The database connection is handled through a utility class named `DBConnection`. The connection class provides a centralized method for creating a connection with PostgreSQL.

The general database connectivity process is:

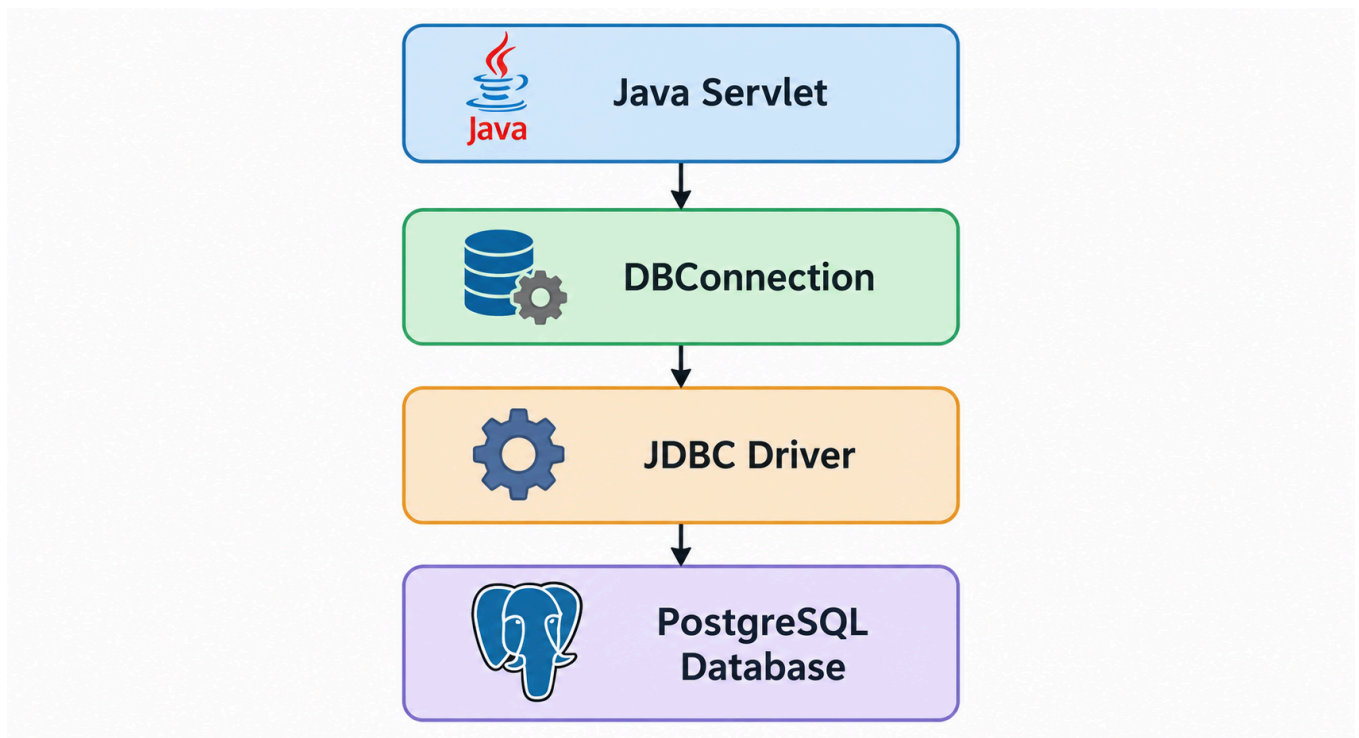


Figure 3.10: Java Servlet to PostgreSQL Database Connectivity Flow Using JDBC

The application uses SQL statements to perform database operations. Prepared statements can be used to execute queries with user-provided values and improve input handling.

Database connectivity is required for:

- Candidate registration.
- Recruiter authentication.
- Job management.
- Application management.
- Interview scheduling.
- Talent matching.
- Analytics.

3.13 BACKEND IMPLEMENTATION

The backend of the system is implemented using Java Servlets. Servlets process requests received from the frontend and perform the required business logic and database operations.

The project contains separate servlet classes for different recruitment functions.

AuthServlet

`AuthServlet` handles authentication-related operations such as recruiter login.

CandidateServlet

`CandidateServlet` manages candidate-related operations such as candidate registration and candidate information processing.

JobServlet

`JobServlet` manages job-related operations. It processes job requests and communicates with the database to store and retrieve job information.

ApplicationServlet

`ApplicationServlet` handles job application-related operations.

MatchServlet

`MatchServlet` performs talent matching operations by processing candidate and job skill information.

InterviewServlet

`InterviewServlet` handles interview scheduling and interview-related information.

AnalyticsServlet

`AnalyticsServlet` retrieves recruitment-related information and supports the analytics dashboard.

Utility Classes

The project also contains utility classes such as:

- `DBConnection`
- `SkillMatcher`
- `Validator`
- `PasswordUtil`
- `JSON`

These classes support database connectivity, skill matching, validation, password-related processing, and JSON responses.

3.14 FRONTEND IMPLEMENTATION

The frontend provides the interface through which candidates and recruiters interact with the system.

The frontend is developed using HTML, CSS, and JavaScript.

Home Page

The home page introduces the Intelligent Recruitment Platform and provides navigation to the major sections of the application.

Candidate Registration

The candidate registration interface collects candidate information such as name, email, phone number, and skills.

Recruiter Login

The recruiter login page allows authorized recruiters to access recruitment management functionality.

Job Management

The job management interface allows recruiters to create and manage job postings.

Job Listing

The job listing interface displays available job opportunities to candidates.

Job Application

Candidates can submit applications for available jobs through the application interface.

Interview Scheduling

The interview interface allows interview-related information to be managed.

Analytics Dashboard

The analytics page displays recruitment statistics and provides a summarized view of recruitment activities.

3.15 TALENT MATCHING ALGORITHM

The Talent Matching module compares candidate skills with the skills required for a particular job.

The basic matching process is:

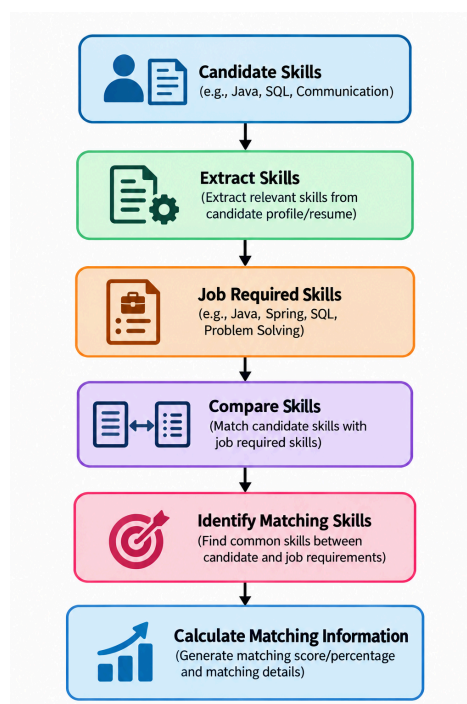


Figure 3.11: Candidate–Job Skill Matching and Matching Information Calculation Process

The matching process can be represented using the following formula:

$$\textbf{Matching Percentage} = (\textbf{Number of Matching Skills} / \textbf{Total Required Skills}) \times 100$$

For example, if a job requires five skills and a candidate has four of those skills:

$$\textbf{Matching Percentage} = (4 / 5) \times 100 = 80\%$$

The matching result helps recruiters identify candidates whose skills are more closely related to the requirements of a job.

The `SkillMatcher` utility class is responsible for supporting the skill comparison process.

3.16 ANALYTICS IMPLEMENTATION

The Analytics Dashboard provides summarized information about recruitment activities.

The analytics module can process information stored in the database and present it in an understandable format.

Important recruitment metrics include:

- Total candidates.
- Total jobs.
- Total applications.
- Scheduled interviews.
- Completed interviews.
- Cancelled interviews.
- Application status information.
- Recruitment activity information.

The analytics process is:

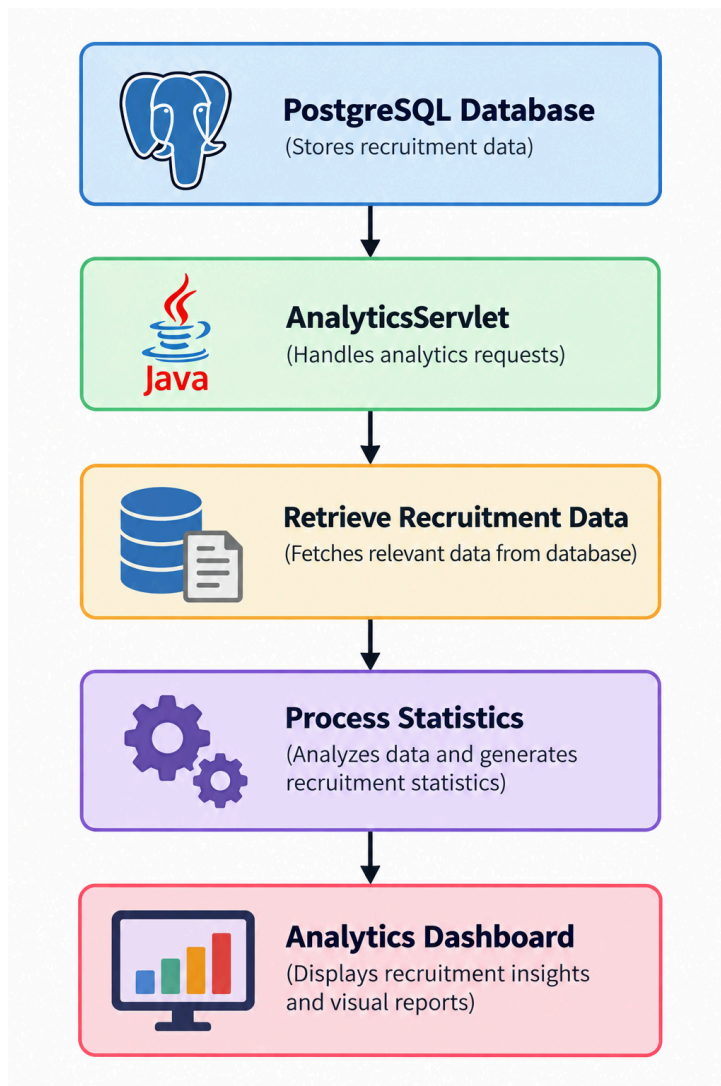


Figure 3.12: Recruitment Data Processing and Analytics Dashboard Generation Flow

The dashboard helps recruiters understand recruitment activity and supports data-driven recruitment decisions.

CHAPTER 4: DATABASE DESIGN

The Intelligent Recruitment and Talent Analytics Platform uses PostgreSQL to store candidate, job, application, and interview information. The database is designed to maintain recruitment records efficiently and ensure data consistency [4].

4.1 CANDIDATE TABLE

The Candidate table stores candidate information including personal details, contact information, educational qualifications, and skills. This information is used during candidate registration, job application, and talent matching processes. The table helps recruiters maintain a centralized repository of candidate profiles and facilitates efficient candidate management.

Field Name	Data Type	Constraints	Description
candidate_id	SERIAL	PRIMARY KEY	Unique Candidate ID
name	VARCHAR(100)	NOT NULL	Candidate Name
email	VARCHAR(100)	UNIQUE, NOT NULL	Candidate Email
phone	VARCHAR(15)	NOT NULL	Contact Number
skills	TEXT	NOT NULL	Candidate Skills

Table 4.1: Candidate Table Structure

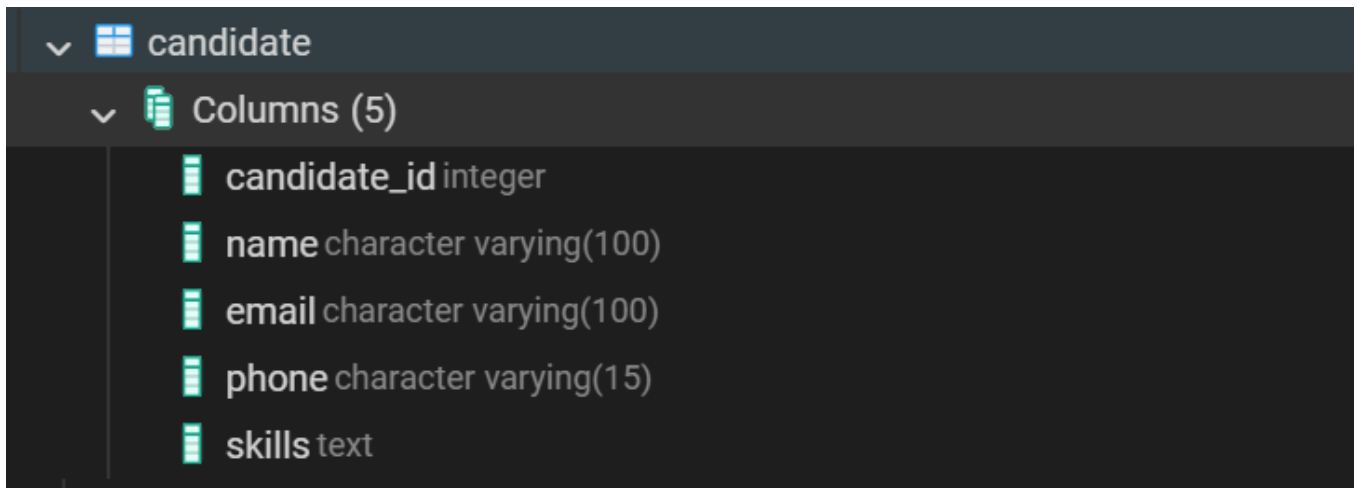


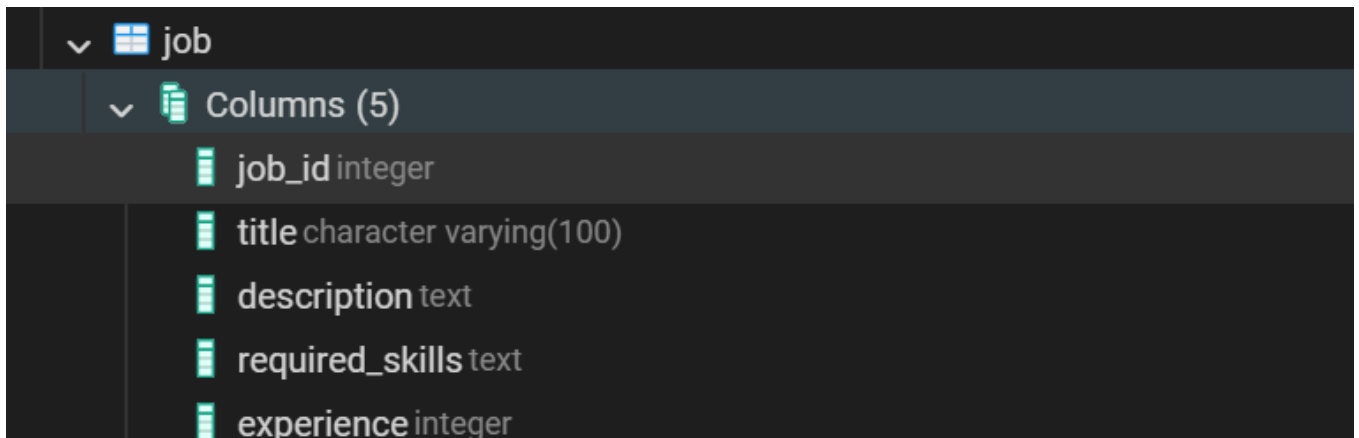
Figure 4.1: Candidate Table in PostgreSQL Database

4.2 JOB TABLE

The Job table stores information related to job openings created by recruiters. It contains details such as job title, job description, required skills, and experience requirements. This table enables recruiters to manage available vacancies and helps candidates identify suitable job opportunities based on their qualifications and skills.

Field Name	Data Type	Constraints	Description
job_id	SERIAL	PRIMARY KEY	Unique Job ID
job_title	VARCHAR(100)	NOT NULL	Job Position
description	TEXT	NOT NULL	Job Description
skills_required	TEXT	NOT NULL	Required Skills
experience	VARCHAR(20)	NOT NULL	Experience Required

Table 4.2.: Job Table Structure



job	
Columns (5)	
job_id	integer
title	character varying(100)
description	text
required_skills	text
experience	integer

Figure 4.2: Job Table in PostgreSQL Database

4.3 APPLICATION TABLE

The Application Table maintains records of job applications submitted by candidates. It establishes a relationship between candidates and job postings by storing application details and status information. This table helps recruiters track applications and monitor candidate progress throughout the recruitment process.

Field Name	Data Type	Constraints	Description
application_id	SERIAL	PRIMARY KEY	Unique Application ID
candidate_id	INT	NOT NULL	Candidate ID
job_id	INT	NOT NULL	Job ID
application_date	DATE	NOT NULL	Date of Application
status	VARCHAR(20)	NOT NULL	Application Status

Table 4.3: Application Table Structure

application	
Columns (4)	
application_id	integer
candidate_id	integer
job_id	integer
status	character varying(30)

Figure 4.3: Application Table in PostgreSQL Database

4.4 INTERVIEW TABLE

The Interview table maintains records of candidate interviews throughout the recruitment process. It stores interview schedules, candidate details, job information, interview dates, and interview status. This table helps recruiters monitor recruitment progress and track interview outcomes effectively.

Field Name	Data Type	Constraints	Description
interview_id	SERIAL	PRIMARY KEY	Unique Interview ID
candidate_name	VARCHAR(100)	NOT NULL	Candidate Name
job_title	VARCHAR(100)	NOT NULL	Job Title
interview_date	DATE	NOT NULL	Interview Date
status	VARCHAR(20)	NOT NULL	Scheduled/Completed/ Cancelled

Table 4.4: Interview Table Structure

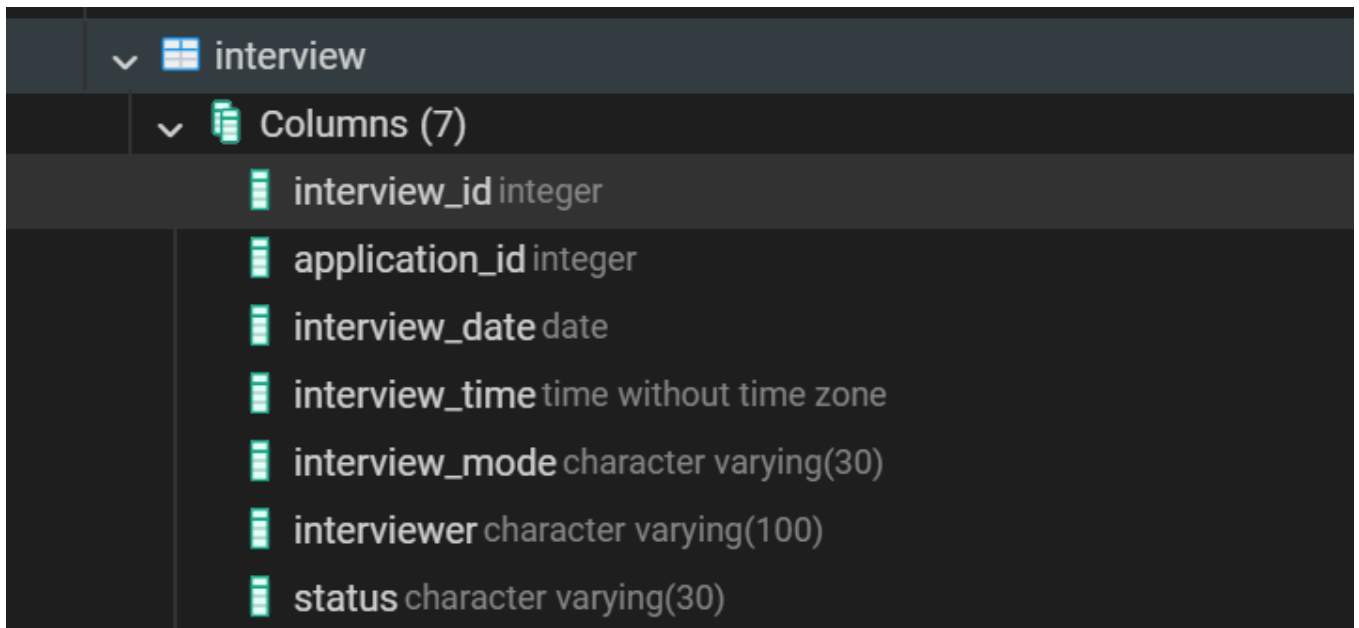


Figure 4.4: Interview Table in PostgreSQL Database

4.5 RECRUITER TABLE

The Recruiter Table stores login credentials and account information for recruiters using the platform. It enables secure authentication and access control, ensuring that only authorized recruiters can manage job postings, candidate applications, interview schedules, and recruitment analytics.

Field Name	Data Type	Constraints	Description
recruiter_id	SERIAL	PRIMARY KEY	Unique Recruiter ID
username	VARCHAR(50)	UNIQUE, NOT NULL	Username
password	VARCHAR(100)	NOT NULL	Password

Table 4.5: Recruiter Table Structure

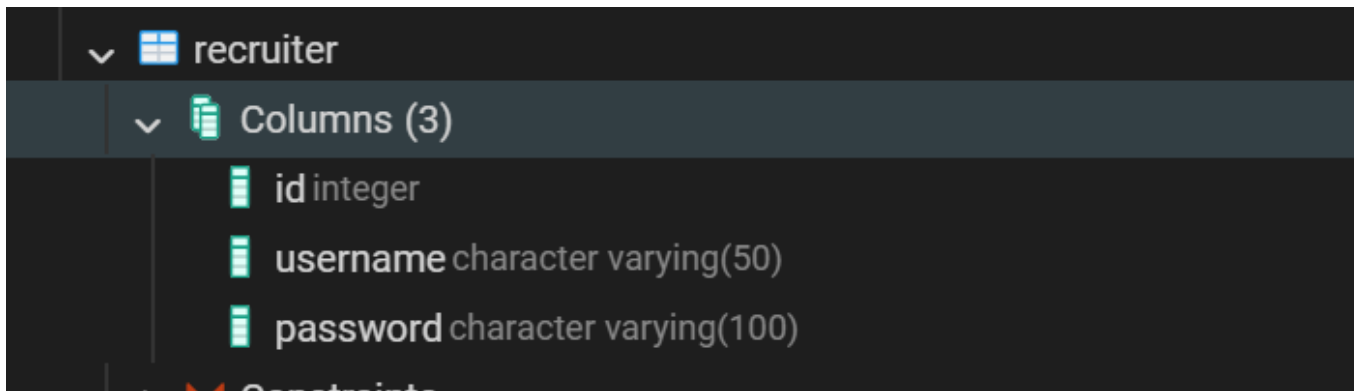


Figure 4.5: Recruiter Table in PostgreSQL Database

4.6 DATABASE RELATIONSHIPS

The database contains multiple entities that are related to one another.

Candidate and Application

A candidate can submit applications for available jobs. Therefore, one candidate can have multiple application records.

Job and Application

A job can receive applications from multiple candidates. The Application table connects candidates with jobs.

Candidate and Interview

Candidates who progress in the recruitment process can be associated with interview records.

Job and Interview

Interviews are associated with job positions for which candidates have applied.

The major relationships can be represented as:

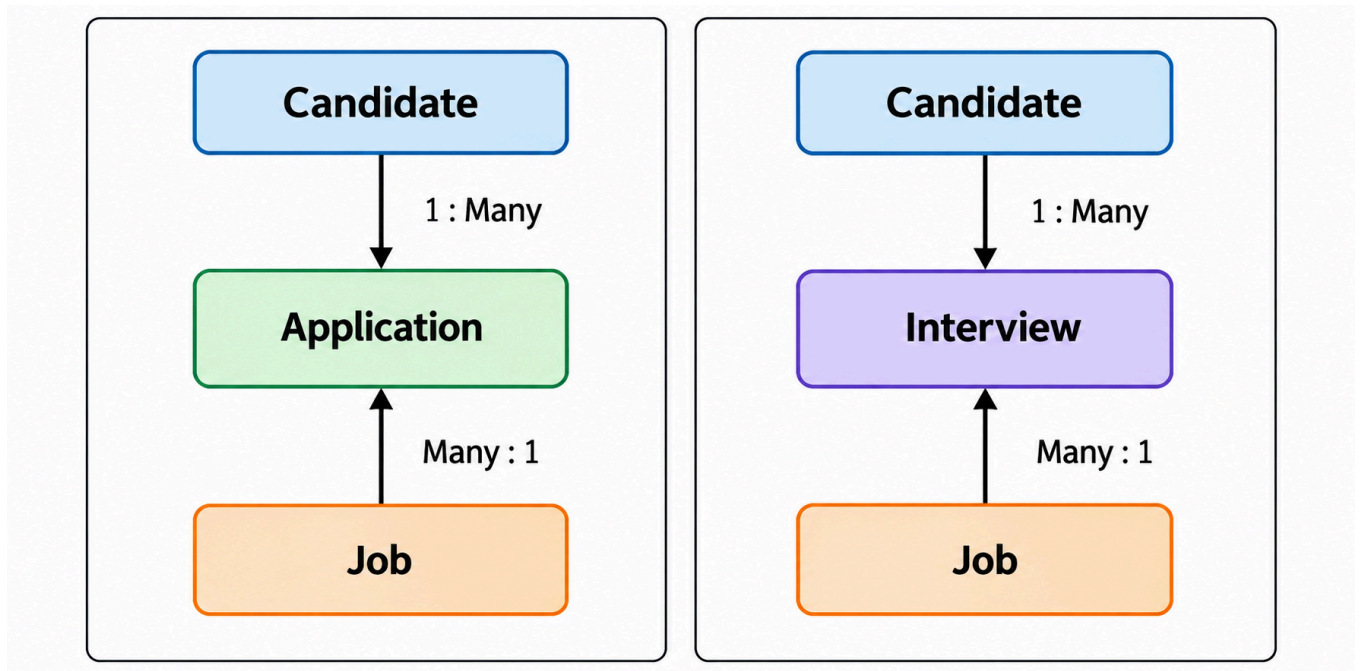


Figure 4.6(a): Candidate–Application–Job Relationship

Figure 4.6(b): Candidate–Interview–Job Relationship

4.7 ENTITY RELATIONSHIP DIAGRAM

The Entity Relationship Diagram represents the main entities of the recruitment platform and their relationships.

The major entities are:

- Candidate
- Recruiter
- Job
- Application
- Interview

Candidate Entity

Stores candidate identification, personal information, contact details, and skills.

Recruiter Entity

Stores recruiter login and authentication information.

Job Entity

Stores job title, description, required skills, and experience requirements.

Application Entity

Connects candidates with job positions and stores application information and status.

Interview Entity

Stores interview-related information including candidate, job, interview date, and status.

The overall relationship structure is:

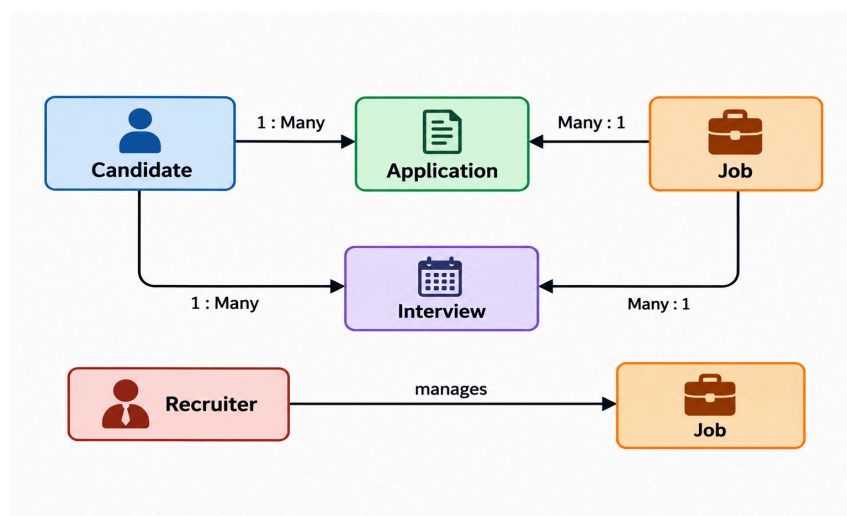


Figure 4.7: Entity Relationship Diagram of Candidate, Application, Interview, Job, and Recruiter

4.8 DATABASE NORMALIZATION

Database normalization is used to organize database information and reduce unnecessary duplication.

First Normal Form (1NF)

Each table stores atomic values, and each field contains a single value. Repeating groups are avoided.

Second Normal Form (2NF)

Tables are organized so that non-key attributes depend on the complete primary key.

Third Normal Form (3NF)

Non-key attributes depend only on the primary key and not on other non-key attributes.

Normalization improves:

- Data consistency.
- Data organization.
- Database maintenance.
- Data integrity.
- Reduction of duplicate information.

4.9 DATABASE CONNECTIVITY

PostgreSQL acts as the central data storage system for the platform. Java Servlets communicate with PostgreSQL through JDBC.

The database connectivity process includes:

1. Loading the PostgreSQL JDBC driver.
2. Establishing a database connection.
3. Creating SQL statements or prepared statements.
4. Executing database operations.
5. Processing query results.
6. Closing database resources.

The database is used by different modules of the system to store and retrieve recruitment information.

CHAPTER 5: RESULTS AND DISCUSSION

The Intelligent Recruitment and Talent Analytics Platform was successfully developed and implemented using HTML, CSS, Java Servlets, JDBC, Apache Tomcat, and PostgreSQL. The system integrates candidate management, job management, talent matching, interview scheduling, and recruitment analytics into a unified platform. All modules were tested successfully and demonstrated proper interaction with the PostgreSQL database.

5.1 RESULTS OBTAINED

The developed system achieved the following outcomes:

- Candidate profiles were successfully registered and stored in the PostgreSQL database.
- Recruiters were able to create, update, and manage job postings efficiently.
- Candidates could browse available job openings and submit applications online.
- Interview schedules were successfully created and maintained.
- Talent matching functionality assisted recruiters in identifying suitable candidates based on skills and job requirements.
- The Analytics Dashboard generated recruitment statistics including interview counts and recruitment progress.
- Data was retrieved and displayed correctly using JDBC connectivity between Java Servlets and PostgreSQL.

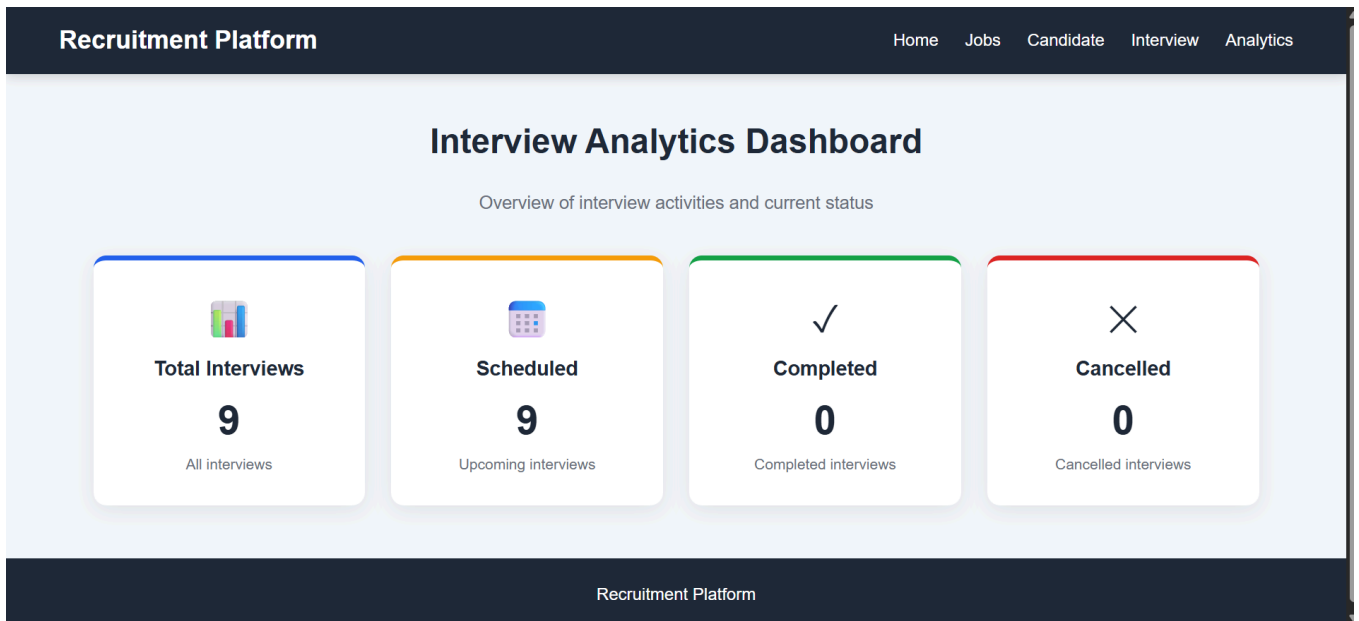


Figure 5.1: Recruitment Analytics Output

5.2 DISCUSSION

The implementation demonstrates how modern web technologies can be utilized to automate recruitment activities effectively. The centralized database improves data organization and eliminates the need for manual record maintenance. The Candidate Management and Job Management modules streamline the recruitment workflow, while the Talent Matching module supports better candidate selection by comparing skills with job requirements [2].

The Interview Scheduling module improves recruitment tracking by maintaining interview records and statuses. The Analytics Dashboard provides valuable insights into recruitment activities, helping recruiters monitor hiring performance and make informed decisions. Overall, the system enhances efficiency, reduces manual effort, and supports structured recruitment management [7].

5.3 ADVANTAGES

The developed platform offers several advantages:

- Centralized recruitment management.
- Reduced paperwork and manual effort.
- Improved candidate-job matching.
- Efficient interview scheduling and tracking.
- Faster access to recruitment information.
- Better recruitment monitoring and reporting.
- User-friendly web-based interface.
- Secure storage of recruitment data using PostgreSQL.
- Improved decision-making through analytics and statistics.

5.4 TESTING RESULTS

Test Case	Expected Result	Status
Candidate Registration	Candidate Stored Successfully	Pass
Recruiter Login	Successful Authentication	Pass
Job Creation	Job Added Successfully	Pass
View Jobs	Jobs Displayed Correctly	Pass
Job Application	Application Submitted Successfully	Pass
Interview Scheduling	Interview Scheduled Successfully	Pass
Talent Matching	Matching Information Generated	Pass
Analytics Dashboard	Statistics Displayed Correctly	Pass

Table 5.1: Testing Results

5.5 FUNCTIONAL TESTING

Functional testing verifies whether the major functions of the system operate according to the expected requirements.

Test Case	Expected Result	Status
Candidate Registration	Candidate details stored successfully	Pass
Recruiter Login	Recruiter authentication completed successfully	Pass
Job Creation	Job added successfully	Pass
View Jobs	Available jobs displayed correctly	Pass
Job Application	Application submitted successfully	Pass
Talent Matching	Matching information generated	Pass
Interview Scheduling	Interview information stored successfully	Pass
Analytics Dashboard	Recruitment statistics displayed	Pass

Table 5.2: Functional Testing Results of the Intelligent Recruitment and Talent Analytics Platform

The functional testing results indicate that the major modules of the platform perform their intended operations.

5.6 DATABASE TESTING

Database testing verifies whether information is correctly stored, retrieved, updated, and maintained in PostgreSQL.

The following database operations were considered:

Insert Testing

Candidate, job, application, recruiter, and interview information is inserted into the corresponding database tables.

Retrieval Testing

Stored records are retrieved and displayed through the application.

Update Testing

Existing records can be modified when required.

Delete Testing

Records can be removed when deletion functionality is provided.

Data Validation

Input information is checked before database operations to reduce invalid data entries.

Database testing helps ensure that the application and PostgreSQL database communicate correctly.

5.7 USER INTERFACE TESTING

User interface testing verifies the usability and appearance of the web application.

The following interfaces were checked:

- Home Page.
- Candidate Registration.

- Recruiter Login.
- Job Management.
- Job Listing.
- Job Application.
- Interview Scheduling.
- Analytics Dashboard.

The interfaces are designed to provide simple navigation and clear interaction between users and the system.

5.8 PERFORMANCE DISCUSSION

The performance of the system depends on the efficiency of the frontend, Java Servlet processing, JDBC communication, and PostgreSQL database operations.

The system performs recruitment operations through database queries and server-side processing. Proper database design and efficient SQL operations help reduce unnecessary processing.

The analytics dashboard retrieves recruitment information from the database and presents summarized information to the recruiter.

For a project-level deployment, the system provides sufficient performance for normal recruitment management activities. For large-scale deployment, additional performance optimization and infrastructure improvements may be required.

5.9 LIMITATIONS

Although the platform provides several recruitment management functions, some limitations remain.

- The talent matching mechanism primarily focuses on skill comparison.
- Advanced artificial intelligence-based candidate ranking is not implemented.
- The system does not provide advanced resume parsing.
- Large-scale enterprise deployment may require additional optimization.
- Advanced notification mechanisms are not included.
- The system requires proper database configuration for successful operation.
- Advanced predictive recruitment analytics are not implemented.
- Security mechanisms can be further improved for production-level deployment.

CHAPTER 6: CONCLUSION AND FUTURE SCOPE

6.1 CONCLUSION

The Intelligent Recruitment and Talent Analytics Platform successfully integrates recruitment management, database systems, and web technologies into a unified application. The system automates candidate registration, job management, application processing, interview scheduling, and recruitment analytics. By reducing manual intervention and improving information accessibility, the platform enhances recruitment efficiency and decision-making [7][8].

The project demonstrates practical implementation of Java Servlets, PostgreSQL, JDBC connectivity, and web development technologies. It serves as an effective solution for modern recruitment management requirements and provides a foundation for future enhancements involving artificial intelligence and predictive analytics [2].

6.2 FUTURE SCOPE

- AI-Based Resume Screening
- Machine Learning Candidate Ranking
- Resume Parsing and Skill Extraction
- Email and SMS Notifications
- Video Interview Integration
- Advanced Recruitment Analytics
- Cloud-Based Deployment
- Integration with Professional Networking Platforms

These features would further improve recruitment efficiency and support intelligent hiring processes [2][10].

6.3 FINAL PROJECT OUTCOME

The final outcome of the project is a web-based Intelligent Recruitment and Talent Analytics Platform that integrates recruitment management activities into a centralized system.

The project successfully demonstrates:

- Candidate information management.
- Recruiter authentication.
- Job posting and management.
- Online job application.
- Talent skill matching.
- Interview scheduling.
- Recruitment analytics.
- PostgreSQL database integration.
- Java Servlet-based backend processing.
- HTML, CSS, and JavaScript-based frontend development.

The developed platform provides a structured approach to recruitment management and demonstrates the practical application of web development, Java programming, database management, and recruitment analytics concepts.

CHAPTER 7: REFERENCES

- [1] S. Kumar and R. Sharma, “Online Recruitment Management System Using Web Technologies,” International Journal of Computer Applications, 2024.
- [2] P. Gupta and A. Verma, “Talent Analytics and Data-Driven Recruitment: A Modern Approach,” International Journal of Advanced Research in Computer Science, 2024.
- [3] Oracle Corporation, “Java Servlet Technology Documentation.”
<https://docs.oracle.com>
- [4] PostgreSQL Global Development Group, “PostgreSQL Documentation.”
<https://www.postgresql.org/docs>
- [5] Apache Software Foundation, “Apache Tomcat Documentation.”
<https://tomcat.apache.org>
- [6] Eclipse Foundation, “Eclipse IDE Documentation.” <https://www.eclipse.org>
- [7] M. Fowler, *Patterns of Enterprise Application Architecture*. Addison-Wesley.
- [8] I. Sommerville, *Software Engineering*, 10th Edition, Pearson.
- [9] R. Pressman and B. Maxim, *Software Engineering: A Practitioner's Approach*, 9th Edition.
- [10] A. Rajput and S. Singh, “Web-Based Recruitment and Applicant Tracking System,” IJITEE, 2024.